Classical Mechanics Chapter 14 Question

(Dated: May 4, 2020)

Collision Theory, also known as the *collision* or *scattering experiment*! Despite only being applicable to objects who's size is comparable to atomic nuclei, it is a very powerful tool for investigating the structure of both atomic and subatomic particles. One would carry out the experiment by firing a stream or projectiles (like electrons or protons), at a target (like an atomic nuclei). By observing the distribution of the scattered projectiles that bounced off the target, one can make conclusions about the interactions between the projectiles and target. Now, any sane individual would be tempted to ask the question, what causes the theory to break down as object become more macroscopic? The answer is simple - because we can see macroscopic objects, there are easier ways to calculate the interactions between the projectiles and target.

Without further ado, let's see what a typical question about Collision Theory from John R. Taylor's *Classical Mechanics* looks like:

THE QUESTION

Often times collision calculations can be more simple if looking at things from a viewpoint corresponding to the center of mass frame. Nevertheless, we make measurements in the lab frame, and so it in necessary to be able to convert back and forth between the two. Consider a situation where an initially stationary target is struck by a more massive projectile as seen from the lab frame, as referenced in Figure 1.

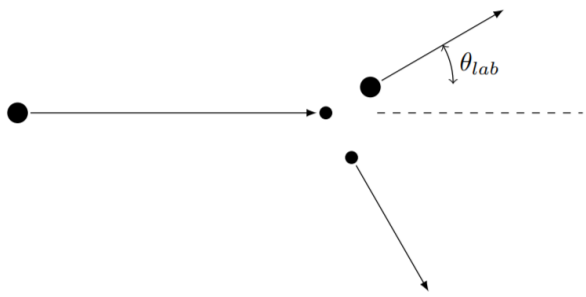

FIG. 1: Visualization of stationary target hit by a projectile with larger mass.

- (a) Draw an analog of Figure 14.15 from your text-book that corresponds to this situation. (Hint: how does λ_P compare to p?) Show that there are two center of mass angles that correspond to each lab angle.

Luckily, Part A simply asks to represent Figure 14.15 in the book (Figure 2), but with the added elements of λ, and a second center of mass angle θ_{CM} that corresponds to θ_{lab}. So what is this lambda - it is simply a mass ratio $\lambda = \frac{m_1}{m_2}$, which in the sketch blow corresponds to a value in which $m_1 > m_2$, thus $\lambda > 1$. Note that λ_P is simply a scaled P, of which is longer than P, placing point A outside of the modeled semicircle, showing that the projectile is more massive than the target. P and P' are simply the initial and final relative momenta, with P'' resembling a second path to D'', also for a projectile. The two destinations D' and D'' grants the creation of two θ_{CM} angles.

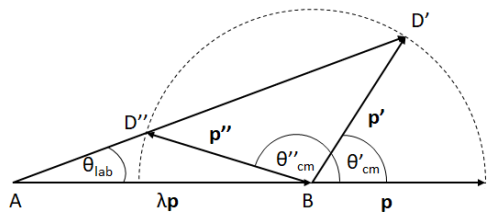

FIG. 2: Analog of figure 14.15 accounting for a more massive projectile. Includes two θ_{CM}, and shows size comparison of P to λ_P.

- (b) What are the two center of mass angles that correspond to $\theta_{lab} = 0$? Explain why there is ambiguity in this particular case.

Considering $\theta_{lab} = 0$, it is easy to see that θ_{CM} would be equal to 0. However, the momentum of the incident projectile must be conserved. While $\theta_{CM} = 0$, it is just as likely that $\theta_{CM} = \pi$, thus providing two different destinations D' and D'', and showing that the magnitude of the momentum is conserved. Note that this is part of a separate condition that John R. Taylor proves through problem 14.25.

- (c) Find an expression for θ_{lab} as a function of θ_{CM} and λ.

Some math is needed to derive equation 14.53, which is exactly what will later be plotted:

$$\theta_{lab} = arctan\frac{sin\theta_{CM}}{\lambda + cos\theta_{CM}}. \tag{1}$$

To do this, imagine a perpendicular line connecting the horizontal, bottom line in the figure from Part A, to D'. Let's name this point F, as seen in Figure 3.

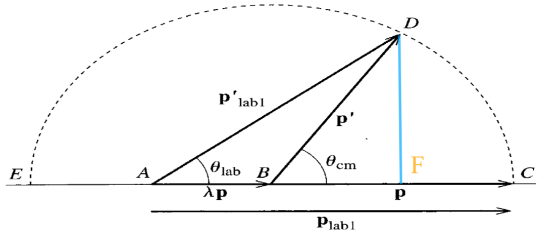

FIG. 3: Example of a line drawn from the edge of the semicircle to the horizontal. NOTE: This is not the exact visual aid representing this problem, rather an example of how the line is created

This will allow θ_{lab} to be written with magnitudes, such that

$$tan\theta_{lab} = \frac{\mid DF \mid}{\mid AF \mid}. \tag{2}$$

Rewriting $\mid DF \mid$ and $\mid AF \mid$ in terms of their trigonometric functions, along with the P and P' coefficients creates

$$\theta_{lab} = \frac{\mid P' \mid sin\theta_{CM}}{\lambda \mid P \mid + \mid P' \mid cos\theta_{CM}}. \tag{3}$$

Drawing back on the conservation of momentum principal presented in Part A Figure 1, we can establish that $\mid P \mid$ is equal to $\mid P' \mid$, thus leading the cancellation in the numerator and both terms in the denominator, producing Equation 1, which is

$$\theta_{lab} = arctan\frac{sin\theta_{CM}}{\lambda + cos\theta_{CM}}$$

- (d) Plot θ_{lab} as a function of θ_{CM} for the particular cases when $\lambda = 2$, 4 and 8.

This can be done by simply plotting Equation 1, derived above, using the code in Figure 4 for multiple values of λ. In the next few plots, all with different λ values, it is evident that the θ_{lab} values decrease in units of radians as λ increases.

The plots also show that the relationship between θ_{lab} and θ_{CM} becomes more symmetrical, meaning θ_{lab} increases and decreases at a more constant rate as λ increases.

```python
import numpy as np
import matplotlib.pyplot as plt
```

```python
lam = 2
theta = np.arange(0,np.pi,0.01)
tLab = np.arctan((np.sin(theta))/(lam + np.cos(theta)))
plt.plot(theta, tLab)
plt.xlabel('θ cm (rad)')
plt.ylabel('θ lab (rad)')
plt.title('Plot of θ lab vs. θ cm, λ=2')
plt.show()
```

FIG. 4: Python code that allows the plotting of θ_{lab} as a function of θ_{CM}. By simply changing the λ values, different plots can be obtained.

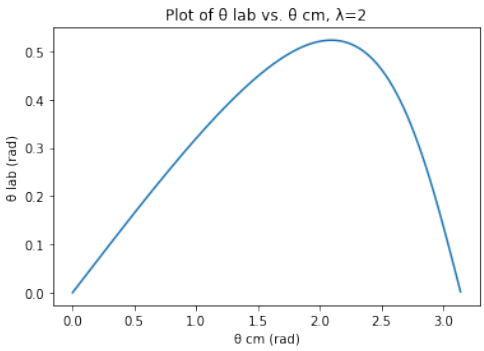

FIG. 5: Graphical relationship between θ_{lab} and θ_{CM} with a λ value of 2.

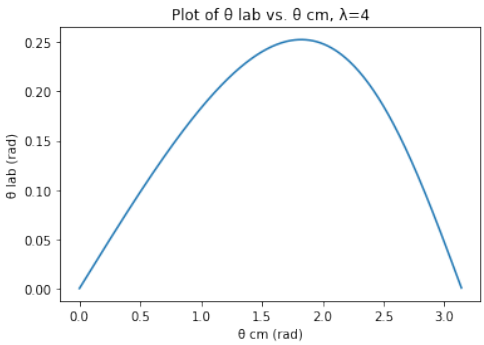

FIG. 6: Graphical relationship between θ_{lab} and θ_{CM} with a λ value of 4.

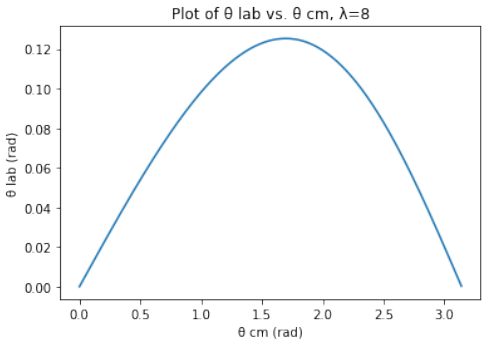

FIG. 7: Graphical relationship between θ_{lab} and θ_{CM} with a λ value of 8.

- (e) Find an expression for the maximum possible value of θ_{lab} for a given value of λ. Check that your solution is correct at least for your plotted values of λ.

Viewing Figure 1 from Part A, it is evident that θ_{lab} has a maximum value. For simplification purposes, Figure 8 shows the new visual aid to use while finding an expression for $\theta_{lab(max)}$. Line $\mid AD \mid$ runs tangential to the semicircle, thus providing the largest $\theta_{lab(max)}$ angle while still modeling the collision interaction between the target and projectile.

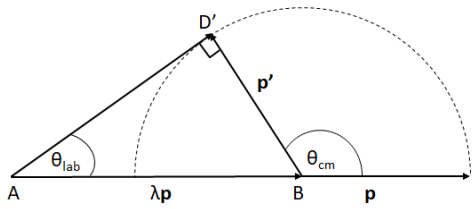

FIG. 8: A new visual showing the scattering of a projectile that obtains the maximum the path along the maximum angle θ_{lab}.

Using Figure 8, it is easy to see that

$$sin\theta_{lab(max)} = \frac{\mid P' \mid}{\mid \lambda P \mid}. \tag{4}$$

As previously established, $\mid P' \mid = \mid P \mid$, thus after simplifying Equation 4,

$$sin\theta_{lab(max)} = \frac{1}{\lambda}. \tag{5}$$

Applying the acrsin to both sides of Equation 5 produces

$$\theta_{lab(max)} = arcsin\frac{1}{\lambda} \tag{6}$$

Equation 6 provides an expression that can calculate the $\theta_{lab(max)}$ for any given λ. Figure 9 shows the substitution $\lambda = 2, 4, 8$ into Equation 6.

```
lam1 = 2
lam2 = 4
lam3 = 8

TMax2 = np.arcsin(1/lam1)
TMax4 = np.arcsin(1/lam2)
TMax8 = np.arcsin(1/lam3)

print('The maximum θ lab when λ=2 is',TMax2)
print('The maximum θ lab when λ=4 is',TMax4)
print('The maximum θ lab when λ=8 is',TMax8)
```

```
The maximum θ lab when λ=2 is 0.5235987755982989
The maximum θ lab when λ=4 is 0.25268025514207865
The maximum θ lab when λ=8 is 0.1253278311680654
```

FIG. 9: substitution $\lambda = 2, 4, 8$ into Equation 6, along with printed solutions below code cell.

$\theta_{lab(max)}$ at $\lambda = 2$ is about 0.523, of which is the same value as the maximum value in Figure 5. Additionally, the other values of $\theta_{lab(max)}$ for various λ values match Figures 6 and 7. Thus, Equation 6 does provide accurate $\theta_{lab(max)}$ values.

(f) Using the fact that $1 + tan^2\theta = sec^2\theta$, Rewrite Equation 1 to be of the form $cos\theta_{lab} = f(cos(\theta_{CM}), \lambda)$.

Equation 1 may be rewritten as

$$tan\theta_{lab} = \frac{sin\theta_{CM}}{\lambda + cos\theta_{CM}}. \tag{7}$$

Square both sides of Equation 7, and rewrite $tan^2\theta$ as

$$1 + sec^2\theta_{lab} = \frac{sin^2\theta}{\lambda^2 + cos^2\theta}. \tag{8}$$

Multiple the denominator of Equation 8 to $1 + sec^2\theta_{lab}$ to produce

$$\lambda^2 + cos^2\theta + \frac{\lambda^2}{cos^2\theta} + 1 = sin^2\theta_{lab}. \tag{9}$$

Knowing that $sin^2\theta + cos^2\theta = 1$, simplify the right side of Equation 9 to obtain

$$\lambda^2 + cos^2\theta + \frac{\lambda^2}{cos^2\theta} + 1 = 1 + cos^2\theta_{lab}. \tag{10}$$

Simplify, and take the square root of Equation 10. Thus,

$$-\lambda - cos\theta_{CM} - \frac{\lambda}{cos\theta_{CM}} = cos\theta_{lab}. \tag{11}$$

Now that Equation 1 is rewritten as $cos\theta_{lab}$ as a function of other cosines and λs, Part f is complete.

(g) Find the lab differential cross-section for a point projectile of mass $2m$ scattering off a hard sphere of radius R and mass m. Plot the lab differential cross section as a function of θ_{lab}.

To begin, it is important to recognize that in order to find the differential cross-section, both the center of mass frame AND the relationship between the center of mass frame to the lab frame are needed. In order to do this, we must take derivatives of the answer from both Part E and Part F to properly account for the two components of the differential cross-section.

The derivative of Equation 5 with respect to $\theta_{lab(max)}$ is

$$cos\theta_{lab(max)} = 0 \tag{12}$$

The derivative of Equation 11 with respect to $cos\theta_{CM}$ is

$$-sin\theta_{CM} - \frac{tan\theta_{CM}}{cos\theta_{CM}} = -sin\theta_{CM} - \frac{sin\theta_{CM}}{cos^2\theta_{CM}} \tag{13}$$

Multiplying Equations 12 and 13 shows

$$-sin\theta_{CM}cos\theta_{lab} - \frac{cos\theta_{lab}sin\theta_{CM}}{cos^2\theta_{CM}} = 1 \tag{14}$$

Solving for θ_{lab} yields

$$sin\theta_{lab} = -\frac{1}{sin\theta_{CM}} - \frac{cos^2\theta_{CM}}{sin\theta_{CM}} \tag{15}$$

Plotting Equation 15 produces Figures 10 and 11. Note: The plot begins after 0 since at 0 Equation 15 is undefined.

```
theta = np.arange(0.1,np.pi,0.1)
tLab = (1/(np.sin(theta))) - np.cos(theta)**2/(np.sin(theta))
plt.plot(theta, tLab)
plt.xlabel('θ lab (rad)')
plt.ylabel('Differential Cross Section')
plt.title('Differential Cross Section as a Function of θ lab')
plt.show()
```

FIG. 10: Code for plotting of Equation 15

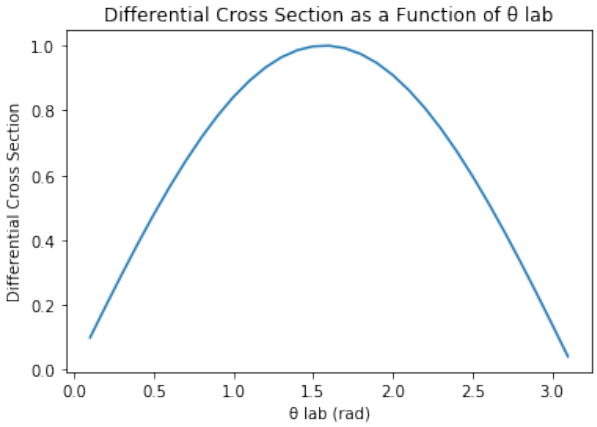

FIG. 11: Plot of Equation 15 against θ_{lab}

WRAP-UP

This question sums up Chapter 14 of John R. Taylor's *Classical Mechanics* quite well. This chapter is especially useful because the subatomic interactions that the Collision Theory models can be visualized quite well, as shown by the figures above. The most difficult part is the concept of Differential Cross-Sections, simply because the components that are derived responsible for interactions between lab and center of mass frames can become quite synonymous, and ultimately blur the clarity of the problem. The interpretation of lifelike events in multiple reference frames is very important to the study of physics as a whole, and the methods outlined in this chapter are sufficient in accounting for the complexity behind the Collision Theory.